BKW

Migration von
MongoDB On-Premise
Clustern in die Cloud

Bern, 21. Juni 2024

BATbern53

Easy Migration through Clean Architecture

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1. Über mich

Hozelito Vajcek

Head of Business Applications Development

TradingUnit@**BKW**

ISAQB® CPSA-A zertifiziert (Advanced Level)

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The Open Group Open Certified

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2. Das Vorhaben - PoC Challenge

PoC: Alle MongoDB On-Premise Cluster in die Cloud migrieren (T - Q - P)

Wie kann sichergestellt werden, dass bei der Migration von On-Premise Servern in die Cloud alle Anforderungen erfüllt werden können und dabei keine „unvorhergesehenen Überraschungen“ auftreten ?

Anforderungen

- ☑ **Sicherheit (Security)**
  - **Private Endpoint** ➔ BKW Azure Infrastruktur
  - MongoDB: *“All data and transportation of Data is encrypted.”*
- ☑ **Zuverlässigkeit (Reliability)**
  - Storage & Instance Size **Auto Scaling**
  - Verfügbarkeit ist gegeben durch die **SLAs der Cloud Anbieter**
- ☑ **ZERO Downtime Approach**
  - Live Migration Funktionalität
- **Performance**
  - **Eigene Lösung (dazu später mehr)**

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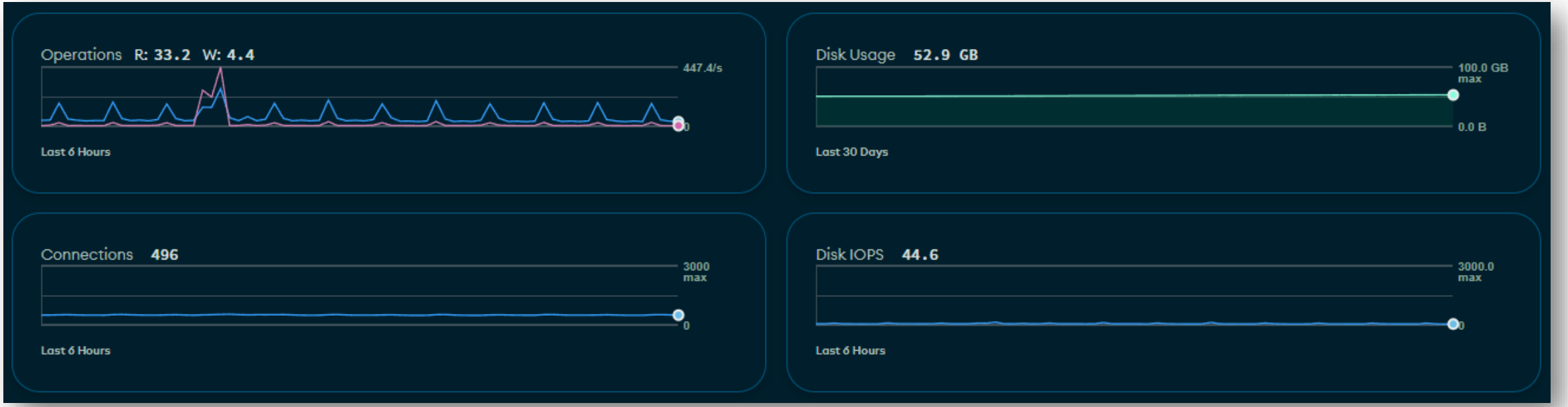
4. MongoDB - Performance Monitoring

OnBoard Monitoring von MongoDB war nicht ausreichend für unsere Zwecke

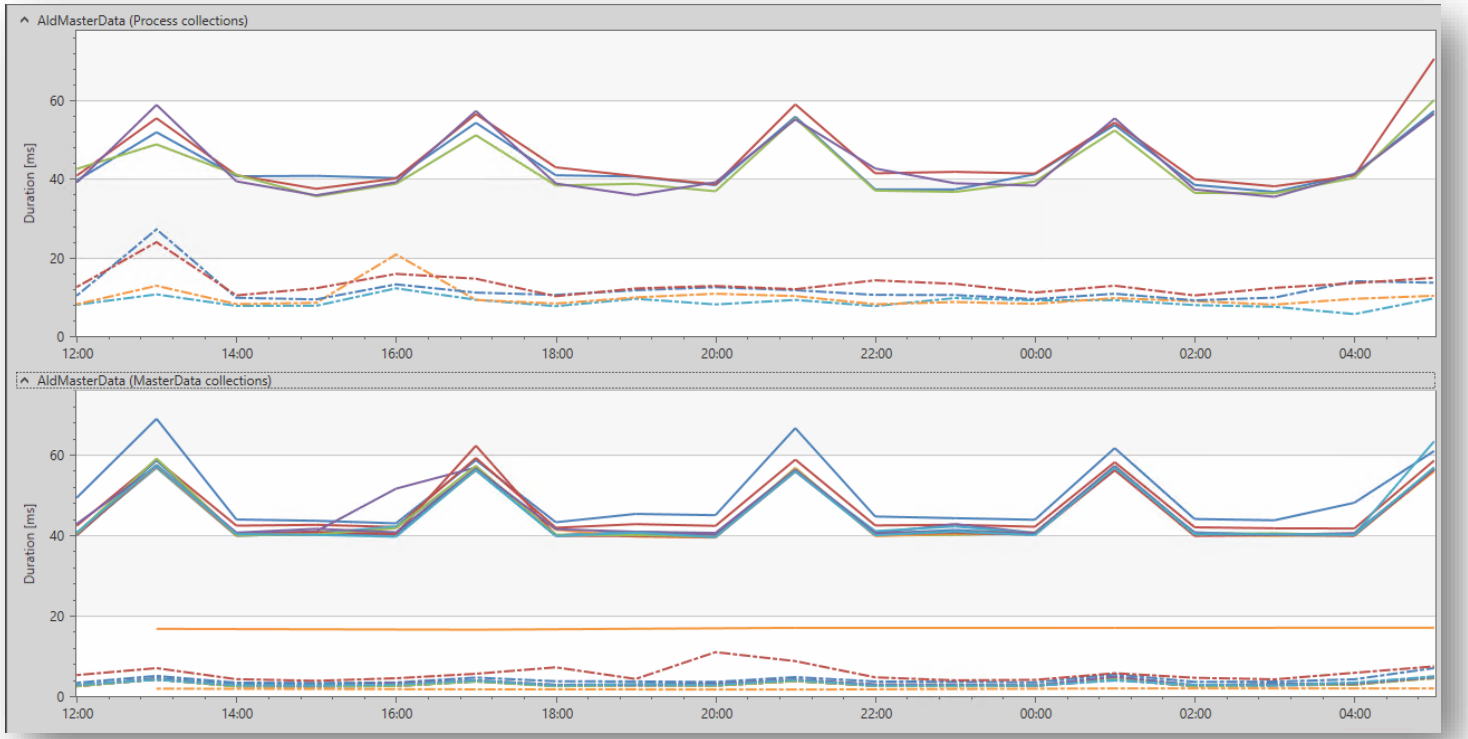
- Isoliertes Monitoring: Lediglich der MongoDB Cluster bzw. die Performance der einzelnen Instanzen werden gemessen
- Kein E2E Monitoring, somit keine Berücksichtigung der Latenzen durch Load Balancer, VPN Tunnel, Proxies, Firewalls etc.



4. MongoDB - Real Performance Monitoring

Anforderungen an das „Real Performance Monitoring“

- E2E Monitoring und somit Messung der echten Performance
- Performance aller Datenbankoperationen mit der Performance nach der geplanten Migration in die Cloud vergleichen und visualisieren



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4. Software Architektur - .NET Services - **CBA**

CBA - Code Bricks Architecture

- .NET Services lassen sich wie Legobausätze zusammenstecken
  - Jeder Service setzt auf ein Framework (Small Bricks) & diversen Komponenten auf
  - Es existieren fertige Komponenten für unterschiedliche Funktionalitäten (Medium Bricks)
  - Gemeinsame Funktionalitäten bei dedizierten Services befinden sich in Common Solutions (Large Bricks)

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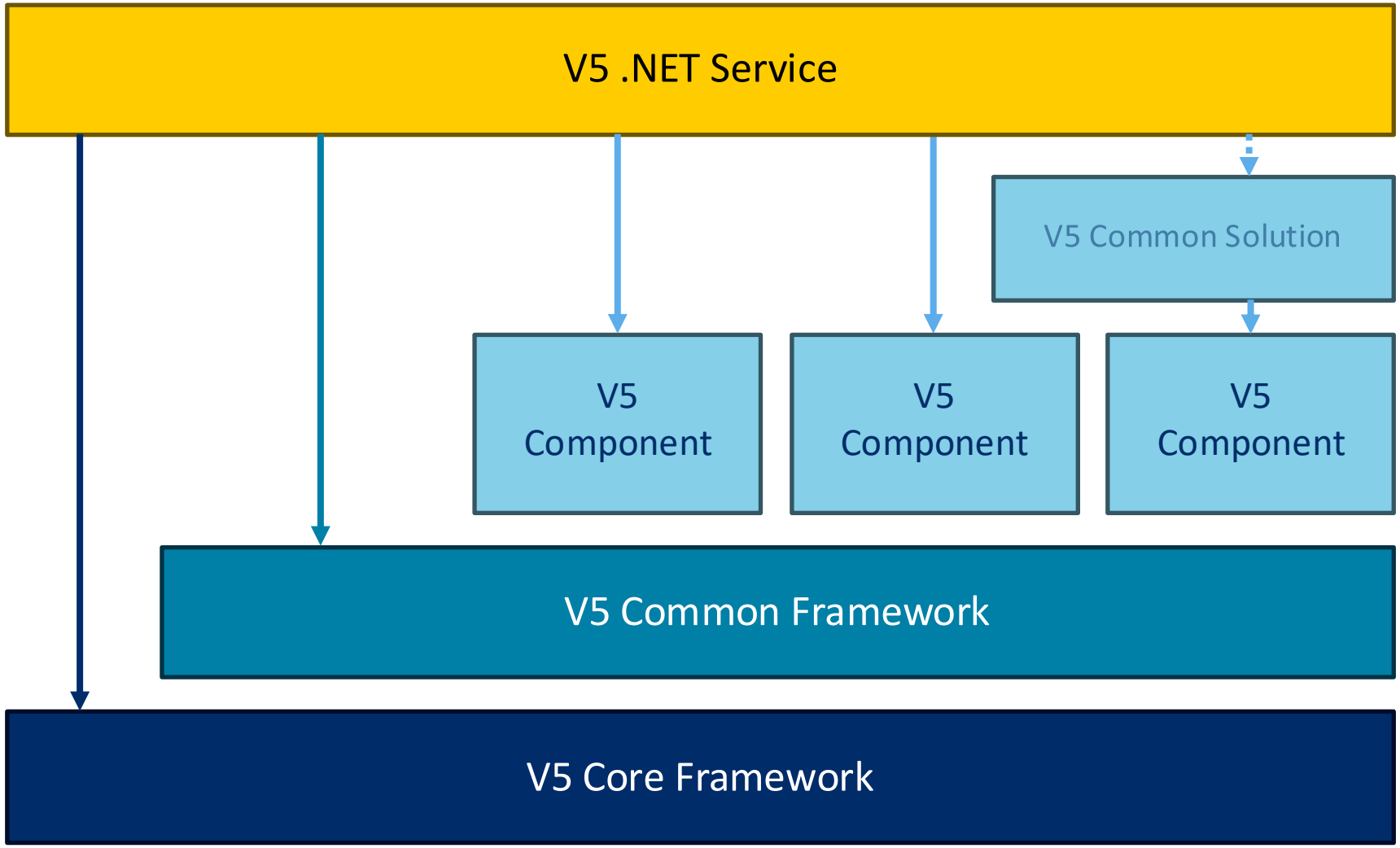
4. Software Architektur - .NET Services - **CBA - Deep Dive**



Windows Service

Based on .NET Core 6

Message Bus:
RabbitMQ

Persistance Layer:

MongoDB

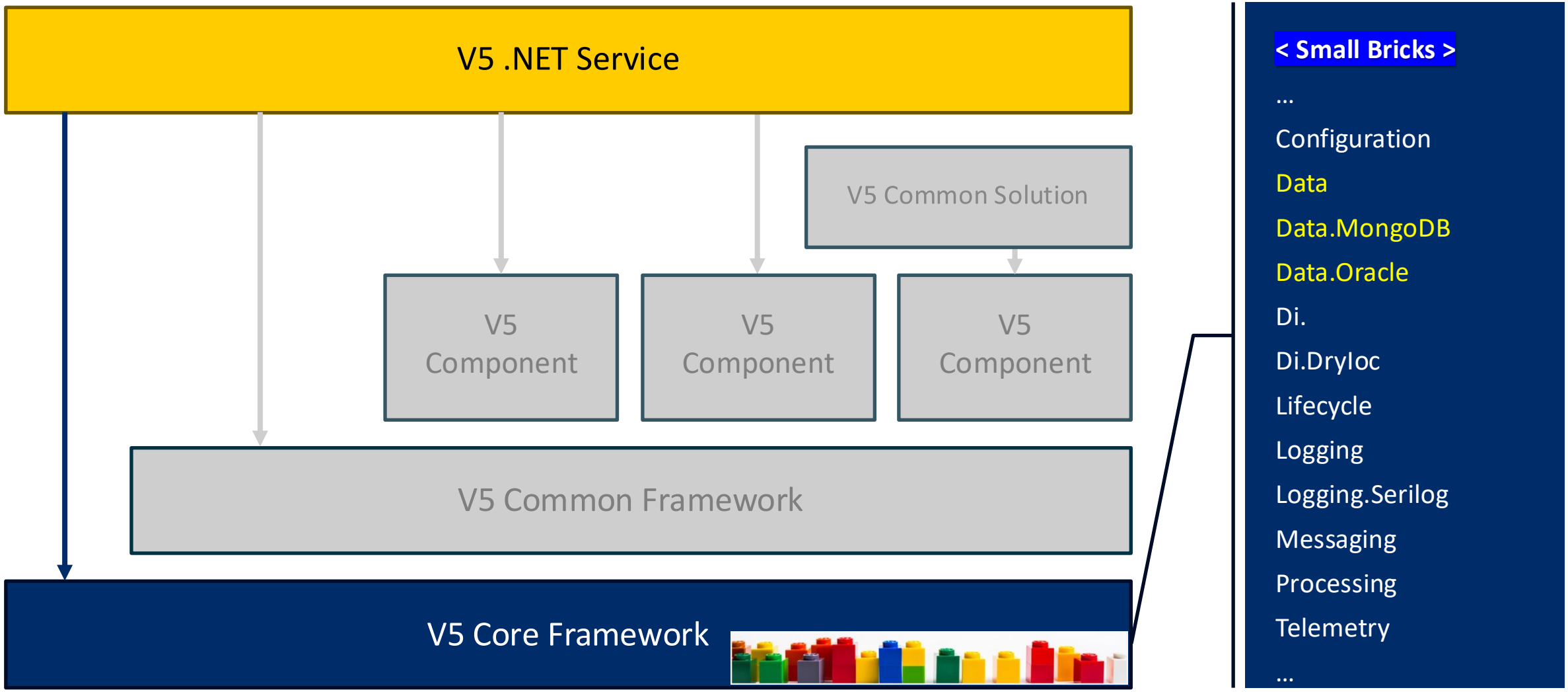
Observability Components:

Loki

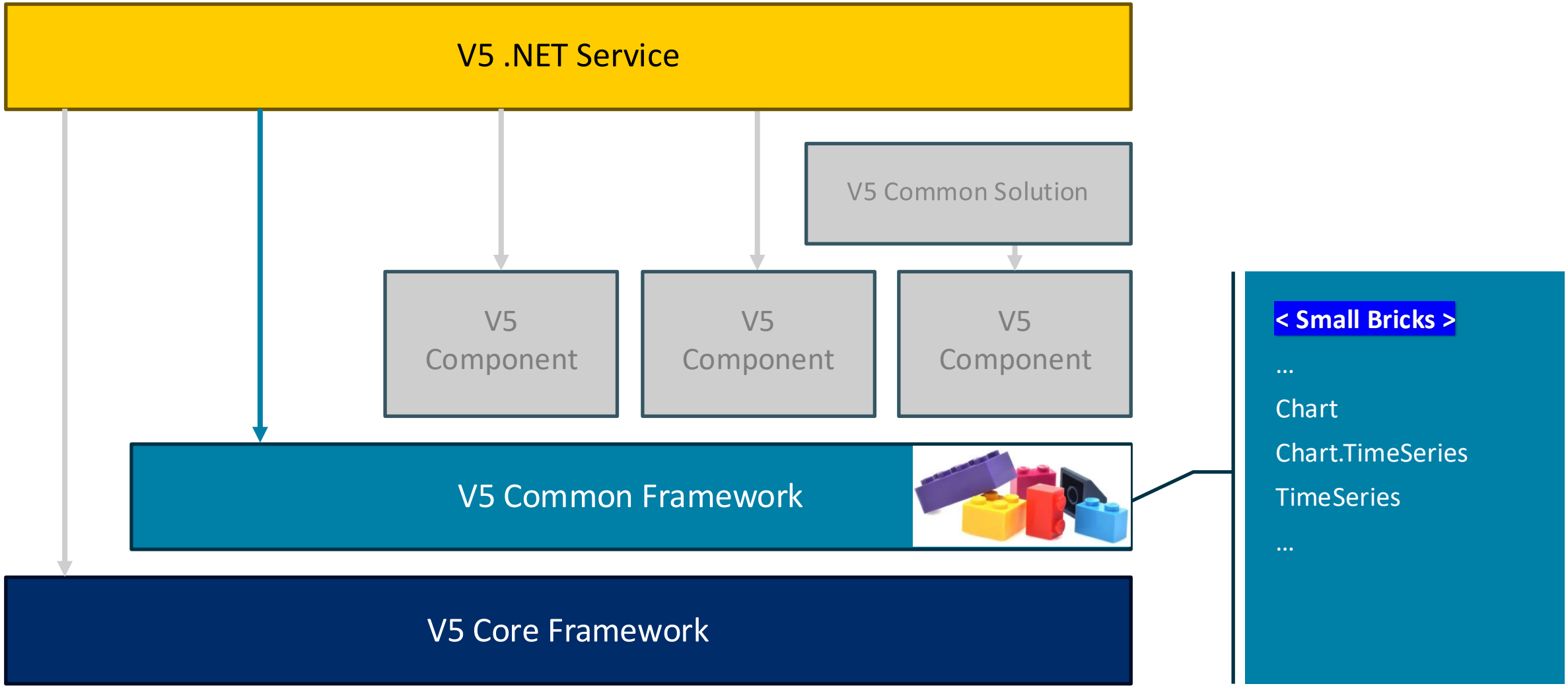
Grafana

Tempo Tracing

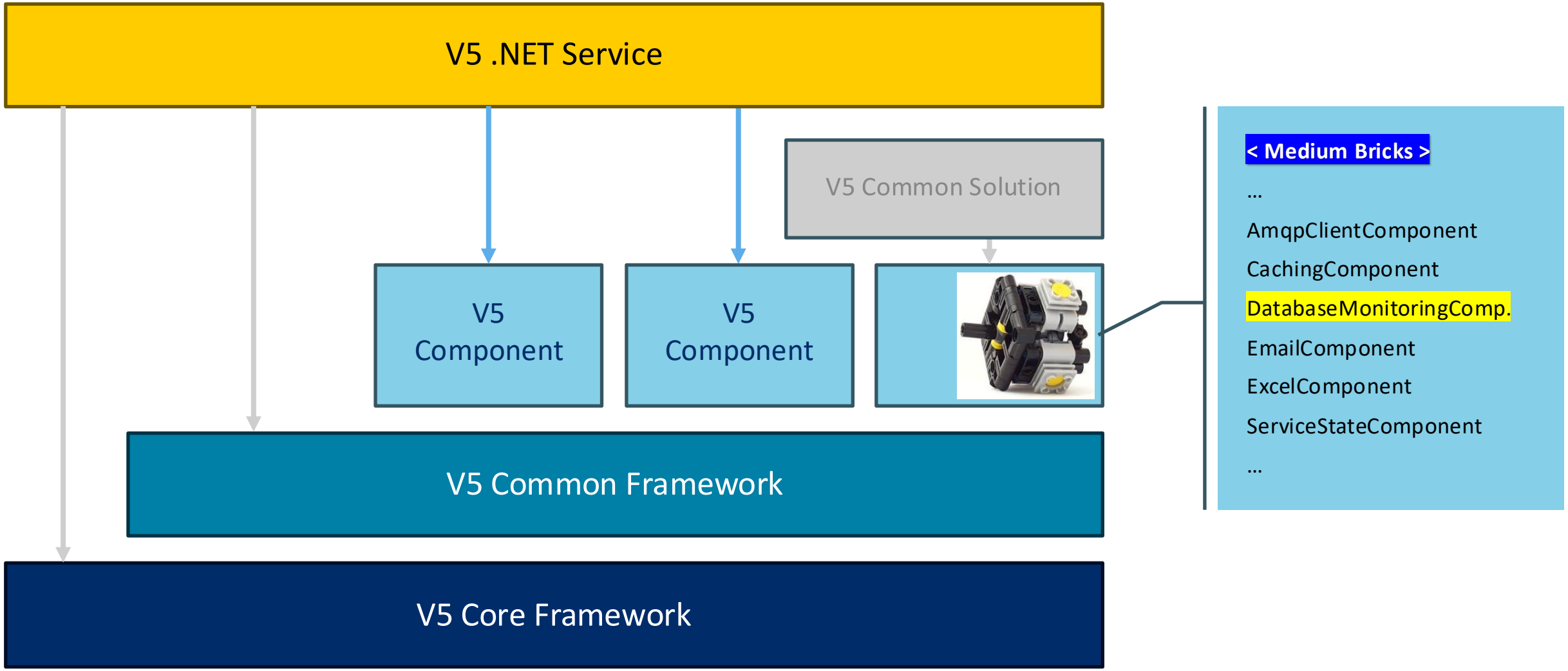
4. Software Architektur - .NET Services - **CBA - Deep Dive**



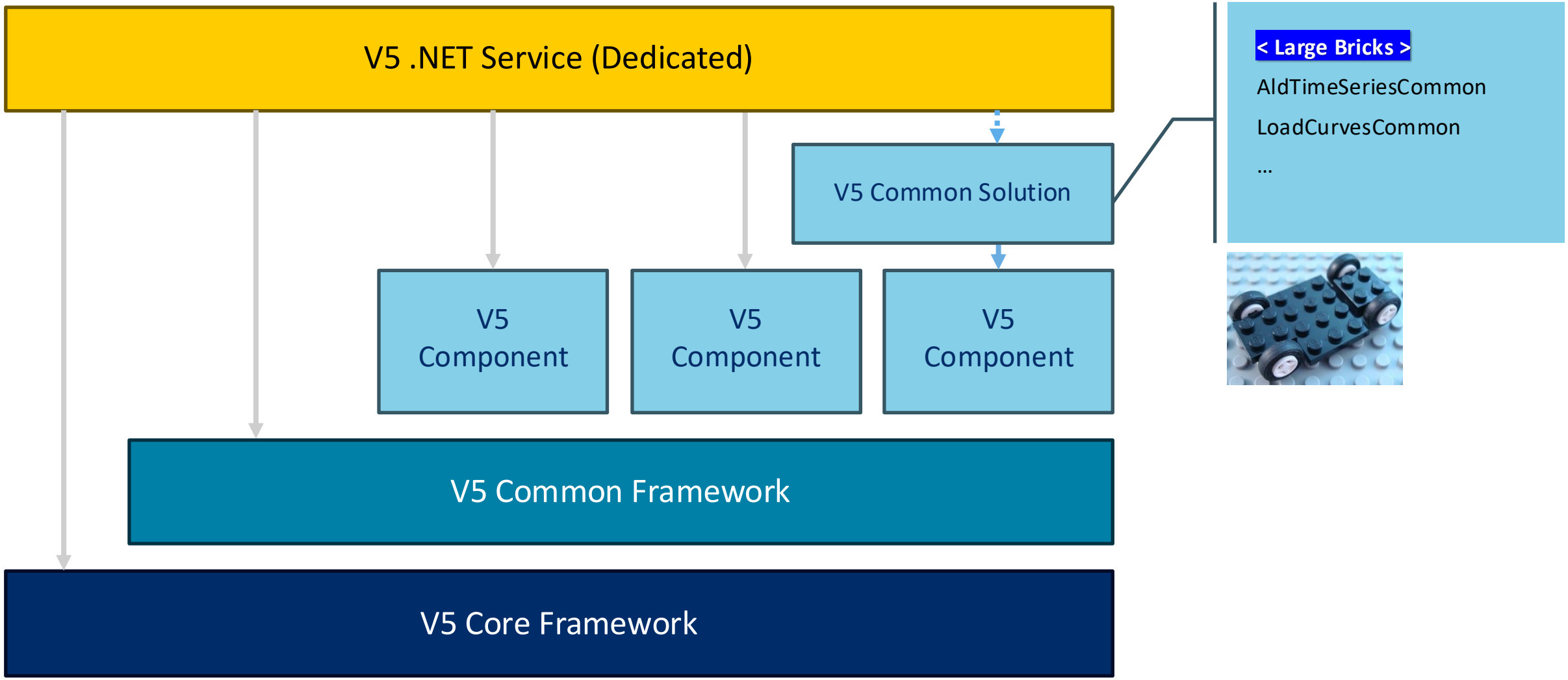
4. Software Architektur - .NET Services - **CBA - Deep Dive**



4. Software Architektur - .NET Services - **CBA - Deep Dive**



4. Software Architektur - .NET Services - **CBA - Deep Dive**



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5. Real Performance Monitoring - Step by Step

Wie können wir die Performance Messung aller Datenbankoperationen in allen .NET Services ermöglichen ?

Backend

- ☑ V5 Core Framework anpassen
- ☑ DB Monitoring Komponente implementieren
  - ☑ DB Performancedaten persistieren
  - ☑ DB Performance Informationen loggen
- ☑ Dedizierte .NET Services zum Testen der neuen Funktionalität umstellen
  - Service A
  - Service B
  - …

Frontend

- ☑ WPF Client für die Anzeige der Performance Daten
- ☑ Vergleich von identischen Zeiträumen visualisieren (1-3 Wochen Diff.)

5. Real Performance Monitoring - Step 1

Wie können wir die Performance Messung aller Datenbankoperationen in allen .NET Services ermöglichen ?

☑ V5 Core Framework anpassen

- Alle MongoDB Operationen als Event anbieten
  - Steuerbar mittels Konfigurations-Schalter im Connection String

Generated with Microsoft Designer Image Creator

```
namespace Bkw.Ad.Core.Data.MongoDb.Implementation;

[LifecyclePriority(1800)]
public class MongoStoreEventProvider<TDataStoreIdentifier> : IDataStoreEventProvider<TDataStoreIdentifier>,
    where TDataStoreIdentifier : IDataStoreIdentifier, new()
{
    Fields

    Constructors

    Properties

    Public Methods

    Private Methods

    #region Events

    public event EventHandler<DataStoreOperationEventArgs> OperationEvent;

    #endregion
}
```

ConnectionStrings:Snowflake	Multiple values	Multiple scopes
	#{MongoDbPrefix}#{Name}#{MongoDbSuffix}	Define scope
	#{MongoDbPrefix}#{Name}#{MongoDbSuffix};triggerEvents=true	Development More...

5. Real Performance Monitoring - Step 2

Wie können wir die Performance Messung aller Datenbankoperationen in allen .NET Services ermöglichen ?

☑ DB Monitoring Komponente implementieren

- MongoDB Core Events sammeln & aggregieren

```
namespace Bkw.Ad.Component.DatabaseMonitoringComponent.Implementation.BackgroundTasks
{
    public class PersistEventsBackgroundTask : IBackgroundTask
    {
        Fields

        Constructors

        #region Public Methods

        public async Task ExecuteAsync(CancellationToken token)
        {
            foreach (var dataStoreOperationEventSubscriber in this.subscribers)
            {
                await dataStoreOperationEventSubscriber.ProcessManyAsync(token);
            }
        }

        #endregion
    }
}
```

```
public override IEnumerable<DatabaseOperationDurationGroup<TDataStore>> CreateDurationsToPersist(IList<DataStoreOperationEvent> events,
{
    events = events.Where(d => d.Duration.TotalMilliseconds >= this.configuration.Current.GroupDurationsInMsAt.Min()).ToList();
    if (!events.Any()) return new List<DatabaseOperationDurationGroup<TDataStore>>();

    var durationGroups = this.GetDurationGroups();
    var durationsToPersist = events
        .GroupBy(
            d => new
                {
                    d.Store,
                    d.Target,
                    d.OperationName,
                    Timestamp = d.Timestamp.ToLocalTime().RoundDown(granularity).ToUniversalTime(),
                    DurationGroup = GetDurationGroup(d.Duration, durationGroups)
                }).Select(
            g => new DatabaseOperationDurationGroup<TDataStore>
                {
                    Id = Guid.NewGuid(),
                    Database = g.Key.Store,
                    Collection = g.Key.Target,
                    OperationName = g.Key.OperationName,
                    Timestamp = g.Key.Timestamp,
                    DurationGroup = g.Key.DurationGroup,
                    Count = g.Count()
                }).ToList();
    return durationsToPersist;
}
```

5. Real Performance Monitoring - Step 3

Wie können wir die Performance Messung aller Datenbankoperationen in allen .NET Services ermöglichen ?

☑ DB Performancedaten persistieren & loggen

- Alle Performancedaten in der MongoDB persistieren und strukturiert loggen

```
private void OnNewEvent(DataStoreOperationEvent operationEvent)
{
    var currentConfiguration = this.configuration.Current;
    if (!currentConfiguration.IsActive || this.ownEntityNames.Contains(operationEvent.Target)) return;

    if (!currentConfiguration.ExcludeTypesFromPersisting.Contains(operationEvent.OperationType))
        this.eventBag.Add(operationEvent);

    if (currentConfiguration.ExcludeTypesFromLog.Contains(operationEvent.OperationType)
        || currentConfiguration.LogIfOperationSlowerThan > operationEvent.Duration) return;

    using (this.log.UseContext("operationResult", operationEvent.OperationResult))
    using (this.log.UseContext("operationId", operationEvent.OperationId))
    using (this.log.UseContext("operation", operationEvent.Operation))
    using (this.log.UseContext("DurationInMilliseconds", operationEvent.Duration.TotalMilliseconds))
    {
        this.log.Log(
            currentConfiguration.DbOperationEventLogLevel,
            "[{PerformanceContext}] {operationName} operation of type {type} executed in {duration}. Database: {database}.
            Constants.PerformanceLogPrefix,
            operationEvent.OperationName.Titleize(),
            operationEvent.OperationType,
            operationEvent.Duration.Humanize(5),
            operationEvent.Store,
            operationEvent.Target);
    }
}
```

- Persistierte Performance Daten -

_internal.Monitoring.DatabaseOperationEvents > OperationType

_id	OperationType	OperationResult	OperationName	Target	Store	Duration	Timestamp	Message
21754143-333E-46E7-BD97-...	2	0	listCollections	null	StreemConnector	00:00:00.0129187	2023-09-17T18:35:47.313Z	null
C5A65AD1-AEAF-4423-A3A...	2	0	listCollections	null	StreemConnector	00:00:00.0133248	2023-09-17T18:35:47.274Z	null
6FA14438-F5F0-40F0-AF0F-...	0	0	find	Installations	StreemConnector	00:00:00.0141029	2023-09-17T18:35:46.133Z	null
3EE61AEB-1358-4CF8-A700-...	1	0	insert	_internal.Audit.MigrationStepExecutions	StreemConnector	00:00:00.0351585	2023-09-17T18:35:46.251Z	null
C9E0C0E3-2D8C-4400-83A1...	0	0	aggregate	GetInstallationsProcessInstances	StreemConnector	00:00:00.0134946	2023-09-17T18:35:47.461Z	null
0DBCABA4-B638-4109-B965...	2	0	listIndexes	GetLoadCurvesProcessInstances	StreemConnector	00:00:00.0136252	2023-09-17T18:35:47.552Z	null
F63B28DB-A571-4019-9B3E-...	2	0	listIndexes	_internal.GetLoadCurvesProcessSnapshots	StreemConnector	00:00:00.0135668	2023-09-17T18:35:47.575Z	null
5F61B34B-00B1-41E4-BCF2-...	2	0	listIndexes	LoadCurvesToGdmProcessInstances	StreemConnector	00:00:00.0137523	2023-09-17T18:35:47.602Z	null
71AE3AC3-B73B-4F58-A3B5...	2	0	listIndexes	_internal.LoadCurvesToGdmProcessSnapshots	StreemConnector	00:00:00.0135706	2023-09-17T18:35:47.641Z	null
E2E87D68-6764-433F-AA6B-...	2	0	listIndexes	Installations	StreemConnector	00:00:00.0135041	2023-09-17T18:35:47.663Z	null
8CC2CAB7-6A02-4C94-998...	0	0	aggregate	GetLoadCurvesProcessInstances	StreemConnector	00:00:00.0135532	2023-09-17T18:35:47.699Z	null
1BEB58FF-603E-4D19-B7EA-...	2	0	listIndexes	GetInstallationsProcessInstances	StreemConnector	00:00:00.0124992	2023-09-17T18:35:47.783Z	null
9570056F-C422-4989-ABA0-...	2	0	listIndexes	GetInstallationsProcessInstances	StreemConnector	00:00:00.0124219	2023-09-17T18:35:47.808Z	null
4771C783-185C-43BC-B8C9...	1	0	update	LoadCurvesToGdmProcessInstances	StreemConnector	00:00:00.0373469	2023-09-17T18:35:48.850Z	null
F139E468-CC49-4123-9F57-...	3	0	isMaster	null	StreemConnector	00:00:00.0138410	2023-09-17T18:35:48.747Z	null
F6BF4C46-46C3-4CC5-9896...	3	0	saslStart	null	StreemConnector	00:00:00.0135720	2023-09-17T18:35:48.774Z	null
FA2DFF4D-233B-4B4B-A52E...	3	0	saslContinue	null	StreemConnector	00:00:00.0135769	2023-09-17T18:35:48.801Z	null
6587F86E-FE13-4407-9C74-...	3	0	isMaster	null	StreemConnector	00:00:00.0134688	2023-09-17T18:35:48.634Z	null
0BD15C49-6F28-4C48-9876-...	3	0	saslStart	null	StreemConnector	00:00:00.0130611	2023-09-17T18:35:48.664Z	null
2012A383-3DB5-4F30-9522-...	3	0	saslContinue	null	StreemConnector	00:00:00.0140270	2023-09-17T18:35:48.692Z	null
7AE3CB0F-399E-4462-BB7A...	2	0	listIndexes	GetInstallationsProcessInstances	StreemConnector	00:00:00.0124534	2023-09-17T18:35:47.839Z	null
C87E0AEE-8D08-4D05-895A...	2	0	dropIndexes	GetInstallationsProcessInstances	StreemConnector	00:00:00.0254758	2023-09-17T18:35:47.902Z	null
7B8CB842-580B-43D6-85C2...	3	0	isMaster	null	StreemConnector	00:00:00.0138238	2023-09-17T18:35:47.401Z	null
1B65768E-D081-40DB-9000-...	3	0	saslStart	null	StreemConnector	00:00:00.0134594	2023-09-17T18:35:47.423Z	null
0687F04E-BE85-43B0-A969-...	3	0	saslContinue	null	StreemConnector	00:00:00.0136250	2023-09-17T18:35:47.450Z	null
C3969CF8-F873-48AE-93E1-...	2	0	listIndexes	GetInstallationsProcessInstances	StreemConnector	00:00:00.0136117	2023-09-17T18:35:47.474Z	null
1833878E-D30F-4738-B0D9-...	2	0	listIndexes	_internal.Audit.MigrationStepExecutions	StreemConnector	00:00:00.0136023	2023-09-17T18:35:47.498Z	null

- Performance Logs -

```
25 Sep 2023 20:12:08.192 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 18 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.173 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 16 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.161 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 17 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.155 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 17 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.150 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 18 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.143 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 17 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.126 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 18 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.119 ● StreemConnector [MongoClient] Sasl Start operation of type Misc executed in 31 milliseconds. Database: StreemConnector. Collection: null
25 Sep 2023 20:12:08.113 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 31 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.107 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 32 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.100 ● StreemConnector [MongoClient] Sasl Start operation of type Misc executed in 38 milliseconds. Database: StreemConnector. Collection: null
25 Sep 2023 20:12:08.094 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 38 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.088 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 39 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.082 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 39 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.075 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 45 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.068 ● StreemConnector [MongoClient] Find operation of type Read executed in 32 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.062 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 38 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.055 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 50 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.049 ● StreemConnector [MongoClient] Find operation of type Read executed in 37 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.043 ● StreemConnector [MongoClient] Find operation of type Read executed in 50 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.036 ● StreemConnector [MongoClient] Find operation of type Read executed in 37 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.030 ● StreemConnector [MongoClient] Find operation of type Read executed in 50 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.024 ● StreemConnector [MongoClient] Find operation of type Read executed in 50 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.017 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 50 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.011 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 52 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:08.005 ● StreemConnector [MongoClient] Find operation of type Read executed in 60 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:07.999 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 62 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:07.992 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 63 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:07.986 ● StreemConnector [MongoClient] Is Master operation of type Misc executed in 55 milliseconds. Database: StreemConnector. Collection: null
25 Sep 2023 20:12:07.980 ● StreemConnector [MongoClient] Find operation of type Read executed in 58 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:07.973 ● StreemConnector [MongoClient] Find operation of type Read executed in 59 milliseconds. Database: StreemConnector. Collection: Installations
25 Sep 2023 20:12:07.967 ● StreemConnector [MongoClient] Find and Modify operation of type Write executed in 59 milliseconds. Database: StreemConnector. Collection: Installations
```

5. Real Performance Monitoring - Step 4

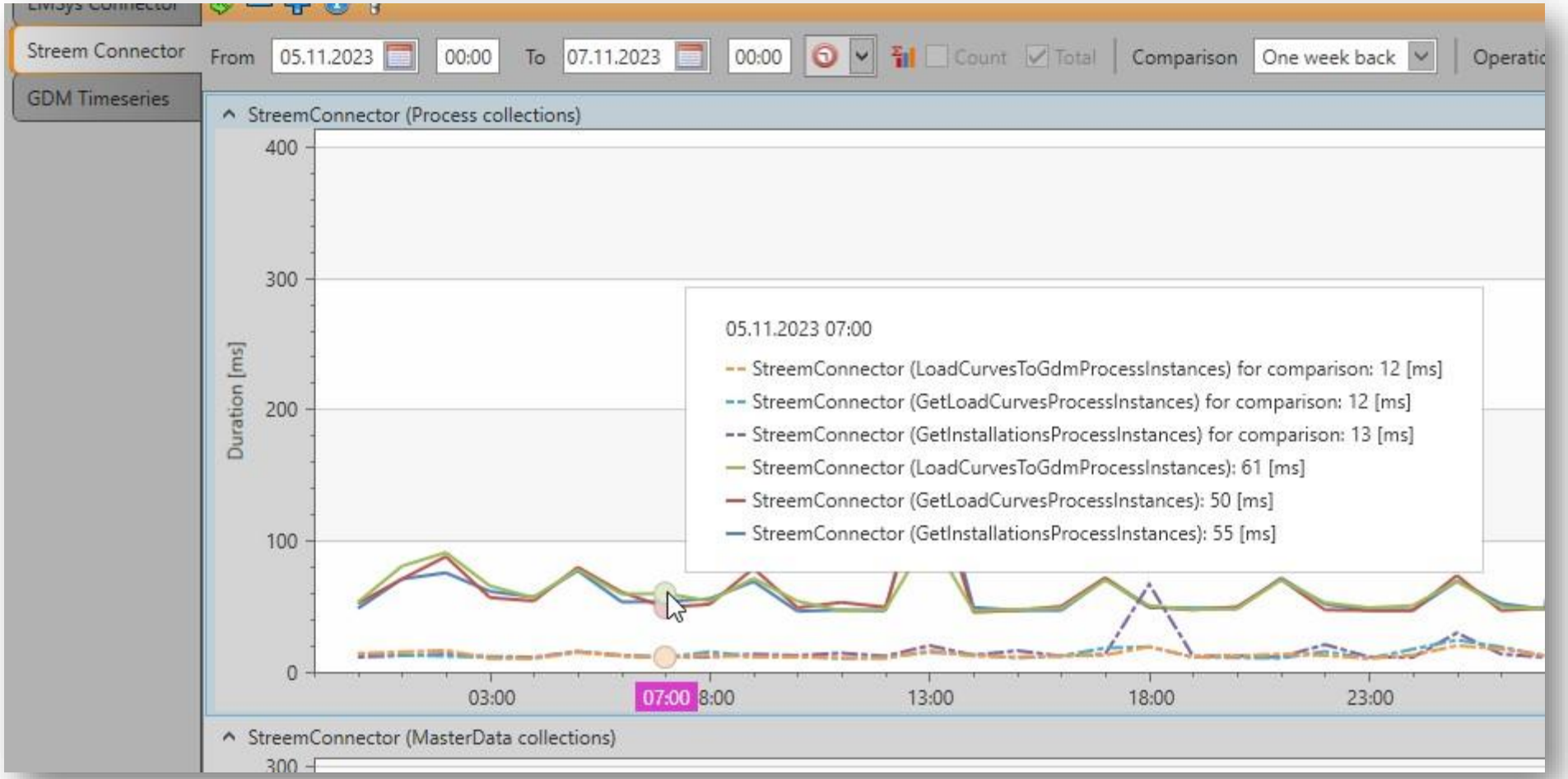
Wie können wir die Performance Messung aller Datenbankoperationen in allen .NET Services ermöglichen ?

- ☑ Dedizierte .NET Services zum Testen der neuen Funktionalität umstellen
  - Connection String anpassen (SPoC - Single Point of Configuration)
- ☑ WPF Client für die Anzeige der Performance Daten
  - ☑ Vergleich von identischen Zeiträumen visualisieren (1-3 Wochen Diff.)

<u>Ablauf des Perfomance Messung</u>

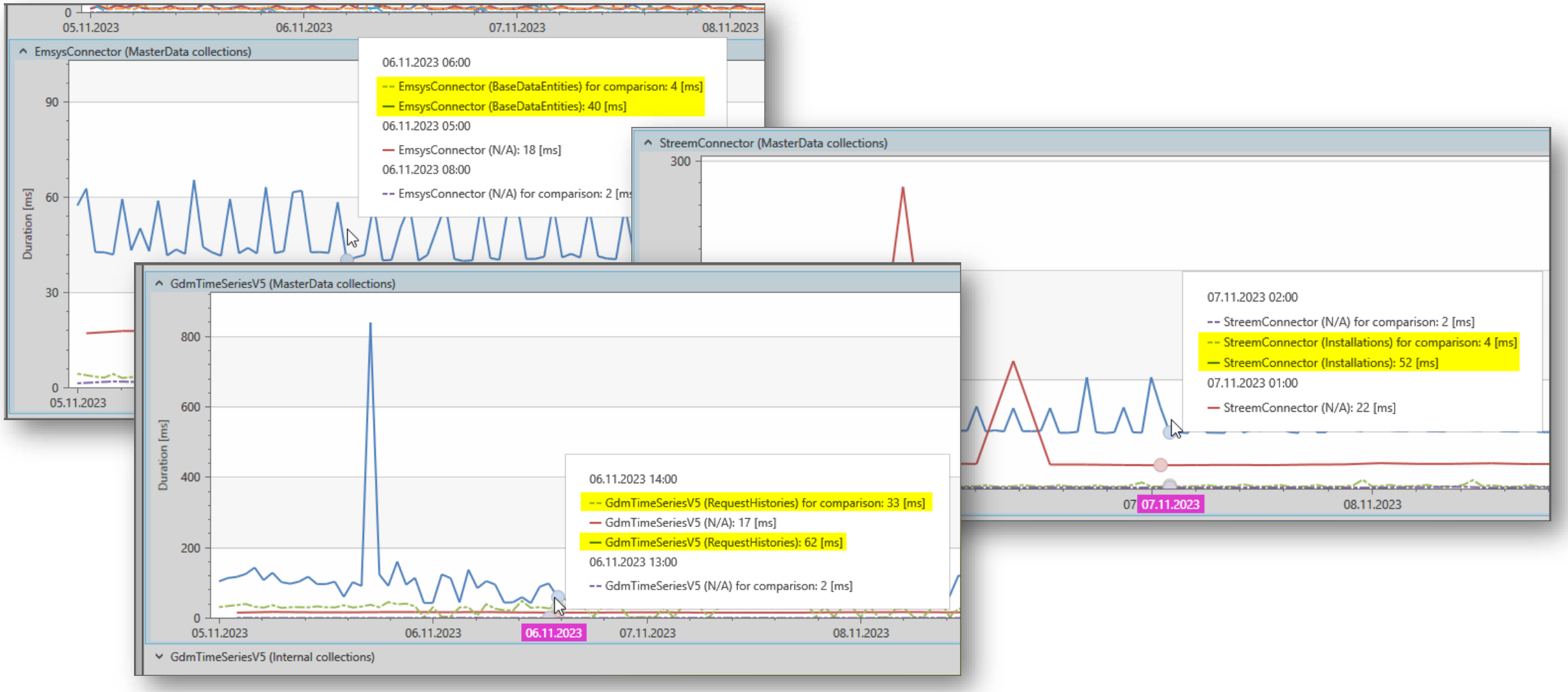
1. Erste 2 Wochen: On-Premise Performance **Datensammlung**
2. Connection String auf den Cloud Cluster umstellen
3. Nächste 2 Wochen: Atlas Cloud Performance **Datensammlung**
4. Performance vergleichen (1 Woche bzw. 2 Wochen zurück)

- Performance Visualisierung -

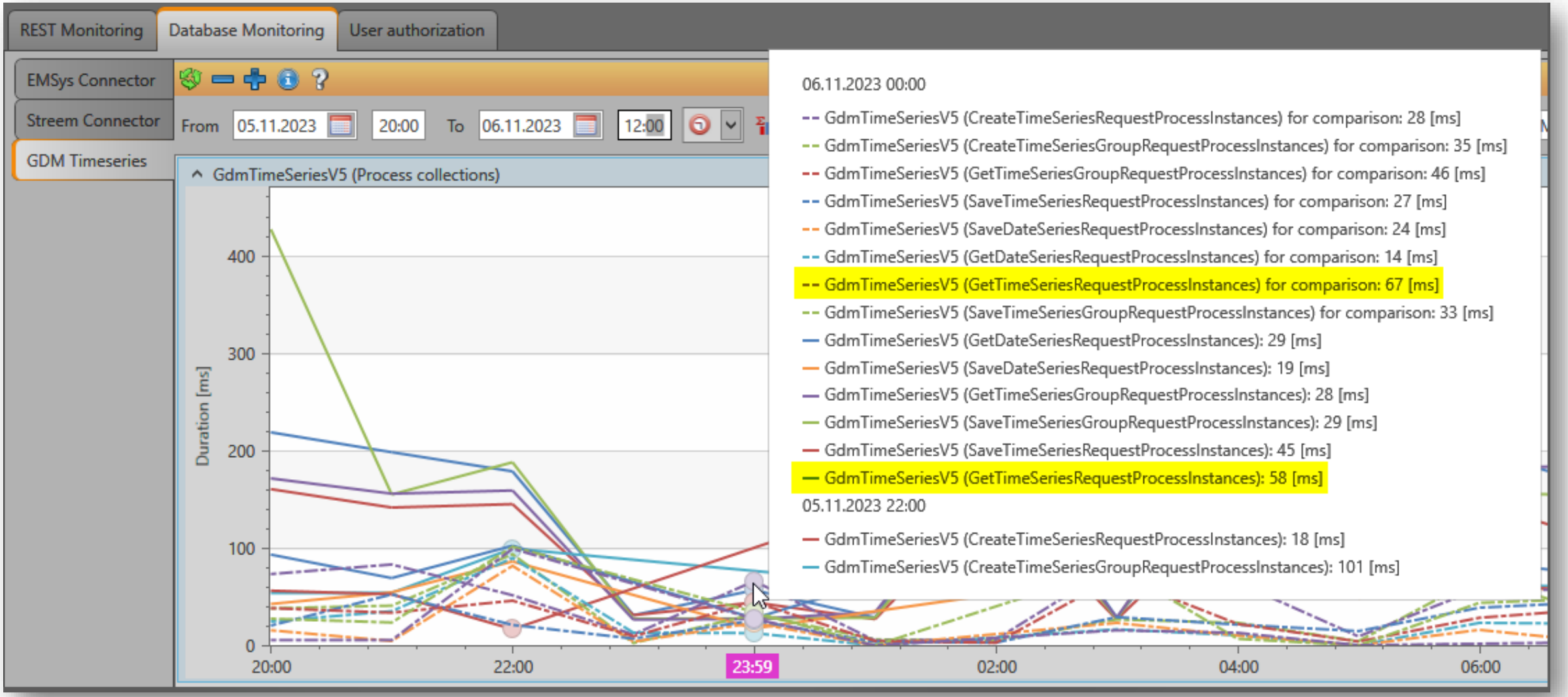


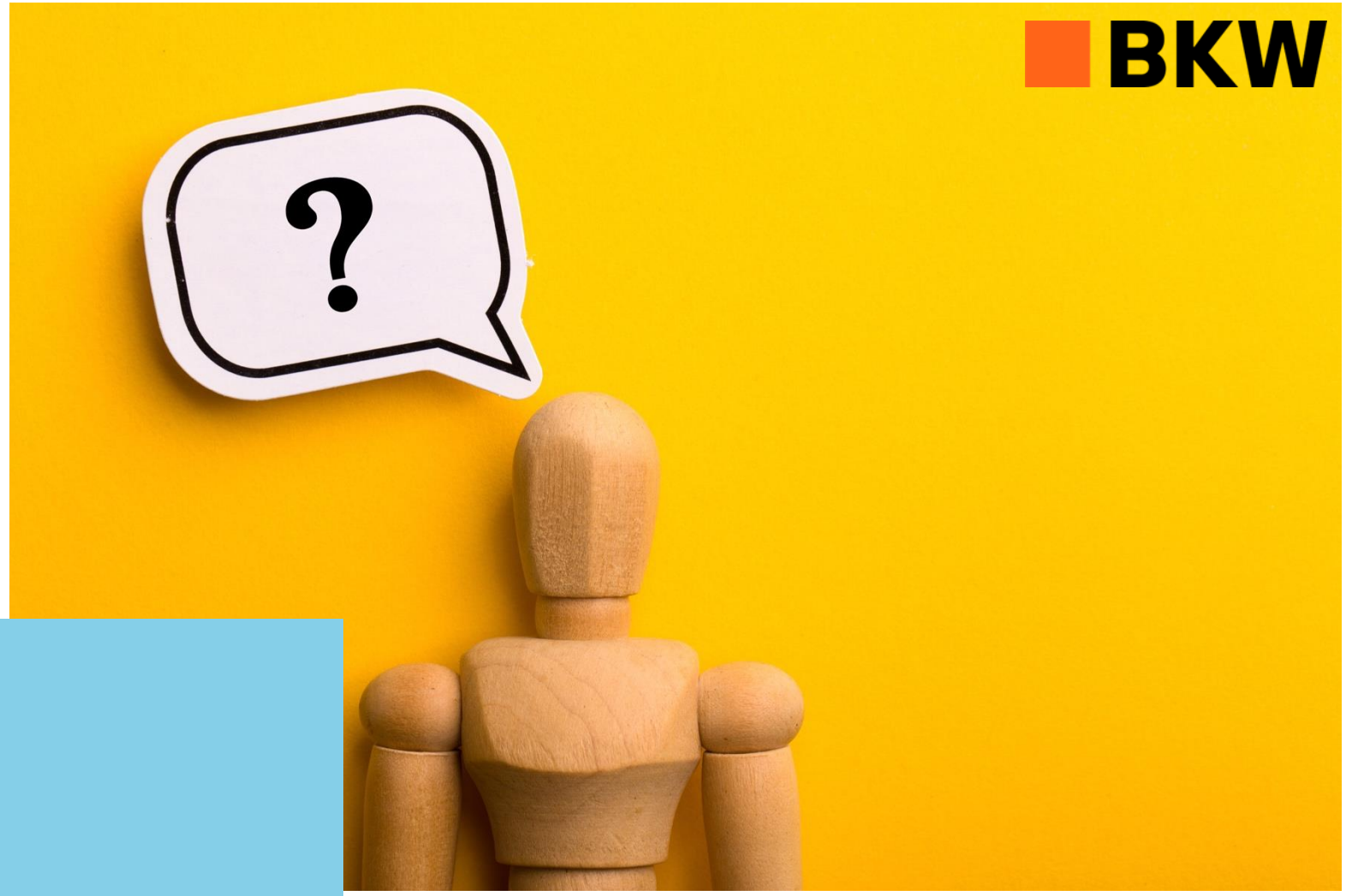
- Performance Visualisierung -

Performanceunterschiede resultieren lediglich aus einer erhöhten Grundlatenz von ca. 30 - 40ms



- Performance Visualisierung -





Q&A

Vielen Dank für Ihre Aufmerksamkeit

Hozelito Vajcek
Head of Business Applications Development